SOLUTIONS

1. $\sqrt[n]{a}$ is called a surd of order n.

2. $5^0 \times 8 = 1 \times 8 = 8$. $[\because 5^0 = 1]$

3. I. $4^1 = 4$ II. $1^4 = 1$
III. $4^0 = 1$ IV. $0^4 = 0$

4. $289 = 17^{\frac{1}{5}x} \Rightarrow 17^2 = 17^{\frac{1}{5}x} \Rightarrow \frac{1}{5}x = 2 \Rightarrow x = 2 \times 5 = 10.$

5. $(81)^4 \div (9)^5 = \frac{(9^2)^4}{(9^5)} = \frac{9^{(2\times 4)}}{9^5} = \frac{9^8}{9^5} = 9^{(8-5)} = 9^3 = 729.$

6. $\left(\frac{9^2 \times 18^4}{3^{16}}\right) = \frac{9^2 \times (9\times 2)^4}{3^{16}} = \frac{(3^2)^2 \times (3^2)^4 \times 2^4}{3^{16}}$

$= \frac{3^4 \times 3^8 \times 2^4}{3^{16}} = \frac{3^{(4+8)} \times 2^4}{3^{16}} = \frac{3^{12} \times 2^4}{3^{16}}$

$= \frac{2^4}{3^{(16-12)}} = \frac{2^4}{3^4} = \frac{16}{81}.$

7. $\frac{4^3 \times 5^4}{4^5} = \frac{5^4}{4^{(5-3)}} = \frac{5^4}{4^2} = \frac{625}{16} = 39.0625.$

8. $\frac{9^3 \times 6^2}{3^3} = \frac{(3^2)^3 \times (3\times 2)^2}{3^3} = \frac{3^{(2\times 3)} \times 3^2 \times 2^2}{3^3} = \frac{3^{(6+2)} \times 2^2}{3^3}$

$= 3^{(8-3)} \times 2^2 = 3^5 \times 2^2 = 243 \times 4 = 972.$

9. $\frac{(19)^{12} \times (19)^8}{(19)^4} = \frac{19^{(12+8)}}{(19)^4} = \frac{(19)^{20}}{(19)^4} = (19)^{(20-4)} = (19)^{16}.$

Hence, missing number = 16.

10. $(64)^4 \div (8)^5 = (8^2)^4 \div (8)^5 = (8)^{(2\times 4)} \div 8^5 = \frac{8^8}{8^5} = 8^{(8-5)} = 8^3.$

11. $(1000)^{12} \div (10)^{30} = \frac{(10^3)^{12}}{(10)^{30}} = \frac{(10)^{(3\times 12)}}{(10)^{30}} = \frac{(10)^{36}}{(10)^{30}} = (10)^{(36-30)}$

$= 10^6 = (10^3)^2 = (1000)^2.$

12. $3^8 \times 3^4 = 3^{(8+4)} = 3^{12} = (3^6)^2 = (729)^2.$

13. $\frac{343 \times 49}{216 \times 16 \times 81} = \frac{7^3 \times 7^2}{6^3 \times 2^4 \times 3^4} = \frac{7^{(3+2)}}{6^3 \times (2\times 3)^4}$

$= \frac{7^5}{6^3 \times 6^4} = \frac{7^5}{6^{(3+4)}} = \frac{7^5}{6^7}.$

14. $\frac{16 \times 32}{9 \times 27 \times 81} = \frac{2^4 \times 2^5}{3^2 \times 3^3 \times 3^4} = \frac{2^{(4+5)}}{3^{(2+3+4)}} = \frac{2^9}{3^9} = \left(\frac{2}{3}\right)^9.$

15. Let $9^3 \times (81)^2 \div (27)^3 = 3^x$. Then,

$3^x = \frac{(3^2)^3 \times (3^4)^2}{(3^3)^3} = \frac{3^{(2\times 3)} \times 3^{(4\times 2)}}{3^{(3\times 3)}} = \frac{3^6 \times 3^8}{3^9} = \frac{3^{(6+8)}}{3^9}$

$\Rightarrow \quad 3^x = \frac{3^{14}}{3^9} = 3^{(14-9)} = 3^5 \Rightarrow x = 5.$

16. Let $6^4 \div (36)^3 \times 216 = 6^{(x-5)}$.
Then, $6^{(x-5)} = 6^4 \div (6^2)^3 \times 6^3$
$= 6^4 \div 6^{(2\times 3)} \times 6^3 = 6^4 \div 6^6 \times 6^3 = 6^{(4-6+3)} = 6$
$\Rightarrow \quad x - 5 = 1 \Rightarrow x = 6.$

17. $(0.2)^2 = 0.2 \times 0.2 = 0.04;$

$\frac{1}{100} = 0.01; \quad (0.01)^{\frac{1}{2}} = [(0.1)^2]^{\frac{1}{2}} = (0.1)^{\left(2\times\frac{1}{2}\right)} = 0.1;$

$(0.008)^{\frac{1}{3}} = [(0.2)^3]^{\frac{1}{3}} = (0.2)^{\left(3\times\frac{1}{3}\right)} = 0.2.$

Clearly, $0.2 > 0.1 > 0.04 > 0.01$.

So, $(0.008)^{\frac{1}{3}}$ is the greatest.

18. $[(2^{-1})^0]^2 = \left[\left(\frac{1}{2}\right)^0\right]^2 = (1)^2 = 1.$

$\left[(4^0)^{-\frac{1}{2}}\right]^2 = \left[(1)^{-\frac{1}{2}}\right]^2 = (1)^2 = 1.$

$\left[(2^{-2})^{-1}\right]^2 = \left[\left(\frac{1}{2^2}\right)^{-1}\right]^2 = \left[\left(\frac{1}{4}\right)^{-1}\right]^2 = (4)^2 = 16.$

$\left[(2^{-1})^2\right]^2 = \left[\left(\frac{1}{2}\right)^2\right]^2 = \left(\frac{1}{4}\right)^2 = \frac{1}{16}.$

19. $(10)^{24} \times (10)^{-21} = 10^{(24-21)} = 10^3 = 1000.$

20. $(256)^{\frac{5}{4}} = (4^4)^{\frac{5}{4}} = 4^{\left(4\times\frac{5}{4}\right)} = 4^5 = 1024.$

21. $(\sqrt{8})^{\frac{1}{3}} = \left(8^{\frac{1}{2}}\right)^{\frac{1}{3}} = 8^{\left(\frac{1}{2}\times\frac{1}{3}\right)}$

$= 8^{\frac{1}{6}} = (2^3)^{\frac{1}{6}} = 2^{\left(3\times\frac{1}{6}\right)} = 2^{\frac{1}{2}} = \sqrt{2}.$

22. $\left(\frac{32}{243}\right)^{-\frac{4}{5}} = \left\{\left(\frac{2}{3}\right)^5\right\}^{-\frac{4}{5}} = \left(\frac{2}{3}\right)^{5\times\frac{(-4)}{5}}$

$= \left(\frac{2}{3}\right)^{(-4)} = \left(\frac{3}{2}\right)^4 = \frac{3^4}{2^4} = \frac{81}{16}.$

23. $\left(-\frac{1}{216}\right)^{-\frac{2}{3}} = \left[\left(-\frac{1}{6}\right)^3\right]^{-\frac{2}{3}} = \left(-\frac{1}{6}\right)^{3\times\frac{(-2)}{3}}$

$= \left(-\frac{1}{6}\right)^{-2} = \frac{1}{\left(-\frac{1}{6}\right)^2} = \frac{1}{\left(\frac{1}{36}\right)} = 36.$

24. $(27)^{-\frac{2}{3}} = (3^3)^{-\frac{2}{3}} = 3^{\left[3\times\left(-\frac{2}{3}\right)\right]} = 3^{-2} = \frac{1}{3^2} = \frac{1}{9}.$

Clearly, $0 < \frac{1}{9} < 1$.

25. $\sqrt[3]{2^4\sqrt{2^{-5}\sqrt{2^6}}} = \sqrt[3]{2^4\sqrt{2^{-5}(2^6)^{\frac{1}{2}}}} = \sqrt[3]{2^4\sqrt{2^{-5}(2)^{\left(6\times\frac{1}{2}\right)}}}$

$= \sqrt[3]{2^4\sqrt{2^{-5}.2^3}} = \sqrt[3]{2^4\sqrt{2^{(-5+3)}}}$

$= \sqrt[3]{2^4\sqrt{2^{(-2)}}} = \sqrt[3]{2^4.(2^{-2})^{\frac{1}{2}}} = \sqrt[3]{2^4.2^{\left(-2\times\frac{1}{2}\right)}}$